

Shuai Li

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Education

Georgia Institute of Technology , Ph.D. Candidate in Operations Research	Aug 2022 - Jun 2027
• Minor: Computer Science	
• Advisor: Dr. Xin Chen and Dr. Weijun Xie	
Fudan University , B.S. in Economics and Minor in Statistics	Aug 2018 - Jun 2022

Research Interests

Theory: Optimization, Approximation Algorithm

Application: Inventory Control, Game Theory and Mechanism Design, Revenue Management

Papers

Faster Decay of the Optimality Gap for Order-Up-To Policies in Lost-Sales Systems with Large Penalty (alphabetic) Xin Chen, Shuai Li , Weijun Xie	Soon to be submitted to MS
High-Probability Integrality of McCormick Relaxations: Beyond Worst-Case Exactness (alphabetic) Xin Chen, Shuai Li , Weijun Xie	Soon to be submitted to OR
The Optimality Gap of Tailored Base-Surge Policies Decays Exponentially in the Regular-Source Lead Time for Dual Sourcing Models (alphabetic) Xin Chen, Shuai Li , Weijun Xie	Submitted to MS
Isotonic Optimization with Fixed Costs (alphabetic) Xin Chen, Shuai Li , Weijun Xie	Major revision at OR
Integrating Market Selection, Inventory Control, and Pricing Decisions (alphabetic) Xin Chen, Shuai Li , Weijun Xie, Haojie Yan	Minor revision at IJO
Joint Assortment Optimization and Marketing Mix Allocation Shuai Li , Zikun Ye, Xin Chen, Weijun Xie	<i>M&SOM</i>
Assortment Optimization in the Presence of Context Effects Reza Yousefi Maragheh, Shuai Li , Tiancheng Zhao, Xin Chen, James Davis, Jason Cho, Sushant Kumar, Kannan Achan	<i>M&SOM</i>

Research Experience

Optimization on Inventory Control and Revenue Management Advised by Dr. Xin Chen and Dr. Weijun Xie	Sept 2022 - present
Importance Sampling for Measure of Rare Event Probability Advised by Dr. Cheng-der Fuh	Jan 2021-June 2022

Internship

Ph.D. Research Aide Technical , Argonne National Lab – Lemont, IL	May 2024 - Aug 2024
Research Scientist Intern , SF Express Co. Ltd – Shenzhen	June 2023 - Aug 2023

Research Assistant in Financial Derivatives, Shenghe International Trade Co., Ltd
– Shanghai

Jul 2020 - Aug 2020

Talks

SIAM Conference on Optimization (Online)	Edinburgh, UK, June 2026
MIP Workshop (Poster Session)	Stamford, CT, May 2026
POMS-HK	Hong Kong, CHN, Jan 2026
INFORMS Annual Meeting	Atlanta, GA, USA, Oct 2025
IISE Annual Meeting	Atlanta, GA, USA, June 2025
INFORMS Annual Meeting	Seattle, WA, USA, Oct 2024
First-year Ph.D. seminar at Georgia Tech	Atlanta, Georgia, USA, 2023

Awards

Inform Student Chapter Spotlight	Georgia Tech 2025
First-year Ph.D. fellowship	Georgia Tech 2022, 2023
Outstanding Student Scholarship	Fudan University 2019, 2021

Academic Service

Journal Reviewer: Naval Research Logistics, Networks

Coursework

- **Graduate:** Linear Optimization, Nonlinear Optimization, Discrete Optimization, Advanced Combinatorial Optimization, Theory of Linear Inequalities, Advanced Integer Optimization, Stochastic Optimization, Design and Analysis of Algorithms, Randomized Algorithm, Stochastic Process I,II
- **Undergraduate:** Regression Analysis, Multivariate Statistical Analysis, Cross Section and Panel Data, Statistical Inference, Time Series Analysis, Nonparametric Statistics, Stochastic Process and Stochastic Analysis, Simulation, Software for Statistics and Econometrics, Real and Functional Analysis, Differential Equation

Skills

Programming Languages: Python (Proficient), C++ (Proficient), R (Proficient), MATLAB (Basic), Stata (Basic)

Languages: Mandarin Chinese, English